

Rebalancing Nature's Scale: A Model to Tame Overtourism

Overtourism has become a global challenge, disrupting economic growth, environmental protection, and residents' quality of life. Rising tourist numbers have caused environmental damage, strained infrastructure, and social tensions. Meanwhile, strict tourism limits can reduce local income and well-being. This dual challenge highlights the need for **sustainable tourism** strategies and data-driven models to optimize resources and maximize overall benefits.

We selected Juneau, Alaska, as the representative city for modeling. Tourist numbers, Equilibrium Line Altitude (ELA), and hotel tax revenue were chosen as indicators for population, environment, and economy, respectively. Inspired by the **Lotka-Volterra model**, **logistic population model**, and **SIS model**, we developed the **Sustainable Tourism Dynamics Model (STDM)**. This system uses ordinary differential equations to capture and visualize interactions among the three dimensions.

Intervention strategies include **tax revenue allocation** and **policies**, enacted by the government. Both consist of predefined measures represented by constraint vectors, serving as impact factors in the **STDM** and evaluation model. Static and dynamic **constraints** ensure the feasibility of these strategies.

According to some materials, we propose to three main factors (**life of local residents**, **local environmental protection**, **local economy and development of society**), and four secondary indicators for each main factor combined with **Analytic Hierarchy Process (AHP)** to attain the comprehensive evaluation score. The values from the strategies and the dynamic system are input to the system, related to the score via various equations.

The model integrates all components into a unified system, where variables strongly connect the sub-models. Intervention strategies serve as inputs to the STDM, and its outputs, combined with strategy choices, determine the **AHP-based evaluation model's score**. This score provides feedback to optimize strategies dynamically. Using this process, governments can adjust interventions and predict their outcomes. For example, our model improves a score from **0.446** to **0.692** through targeted interventions.

We assessed our model's sensitivity using two methods. **The Morris method** identified key factors through the μ - σ diagram, showing that the environmental degradation coefficient (k_2) and tourism revenue factor (k_6) had the strongest direct effects and moderate parameter interactions. **Local sensitivity analysis** revealed that the growth rate of tourist numbers (r_1) and baseline economic growth rate (k_5) were most sensitive, with partial derivatives of **792.18** and **575.2**. In contrast, crowding perception (β_c) and carbon footprint scaling (γ_{CE}) showed low sensitivities, at **0.017** and **-34.5**. These results highlight the importance of high-impact parameters and confirm the model's stability under small changes.

We tested our model's adaptability in Big Sur, a coastal destination facing overtourism. Glacier-related parameters were replaced with region-specific indicators, such as coastal landslide rates ($H_{\text{Sur}} = 2.1$ m/year) and visitor numbers ($N_{\text{Sur}} = 1,532,000$ annually). The recalibrated model improved the normalized sustainability score from **0.32773** to **0.73563** with optimized strategies. Key recommendations included strengthening infrastructure resilience and promoting inland attractions to ease coastal pressure. These results highlight the model's versatility in managing overtourism and supporting sustainable tourism. A memo to the Juneau Tourism Council outlines predictions, impacts of measures, and strategic recommendations for sustainable tourism in Juneau.

Keywords: Overtourism, Sustainable Tourism Dynamic Model, Lotka-Volterra model, SIS model, Intervention Strategies, Analytic Hierarchy Process, Sensitivity Analysis